

ALIAH UNIVERSITY

End-Semester Examination- August, 2021

Year: 2nd Year (Dept. of Civil Engg.)
Subject: Fluid Mechanics & Hydraulics
Full Marks: 80

Semester: 4th Semester
Subject Code: CENUGPC06
Time: 3 hr.

Note: Attempt any 5 questions.

1. Discuss the following terms with suitable sketch and expression if any: **16**
Mass density, Weight density, Specific gravity, Surface tension, Newton's Law of viscosity, Capillarity, gauge and absolute pressure. Steady & unsteady flow, Uniform & non uniform flow, Laminar & turbulent flow, one & multidimensional flow.
2. (a) 1 litre of crude oil weighs $(9.6 + 0.1 \times \text{last digit of R.No})$ N. Calculate its mass density, weight density and specific gravity? **8**
(b) Determine the total pressure on a circular plate of diameter 1.5 m which is placed vertically in water in such a way that centre of plate is 2 m below the free surface of water. Find the position of centre of pressure also. **8**
3. (a) What do you mean by buoyancy and centre of buoyancy? Decipher the condition of equilibrium of a floating and sub merged body. **2+6=8**
(b) A stream function is given by: $\psi = 2x - 5y$. Calculate the velocity component and also magnitude and direction of the resultant velocity at any point. **8**
4. (a) Derive the continuity equation of fluid in Cartesian coordinate system for 3D. Explain local and convective acceleration. **8**
(b) The discharge of water through a pipe of diameter 40 cm is $(400 + 0.1 \times \text{last digit of R.No})$ litre/s. If the pipe is bent by 135° , find the magnitude and direction of the resultant force on the bend. The pressure of flowing water is 29.43 N/cm^2 . **8**
5. (a) Discuss the following terms with suitable sketch and expression if any: **8**
Loss of energy in pipes, Hydraulic gradient and total energy line, Flow through siphon, Flow through parallel pipes, Flow through series pipes, Equivalent pipe and power transmission through pipe.
(b) A pipe of diameter 25 cm and length $(2000 + 0.1 \times \text{last digit of R.No})$ m connects two reservoir, having difference of water level 25 m. Determine the discharge through the pipe. If an additional pipe of diameter 25 cm and length 1000 m is attached to the last 1000 m length of the existing pipe, find the increase in discharge. **8**

6. (a) Discuss the following terms with suitable sketch and expression if any: **8**

Type of similarities, Type of forces acting in moving fluid, Dimensionless numbers and model laws.

- (b) In 1:(30+0.1*last digit of R.No) model of a spillway, the velocity and discharge are 1.5 m/s and 2.0 m³/s; find the corresponding velocity and discharge in the prototype. **8**

Aliah University
Department of Biological Sciences
Even Semester Examination - 2021
Course: 4-Year B-Tech
Semester: IV

Paper Code: BIOUGBS01/ BIO252

Paper Name: Biology for Engineers

Instructions to Candidates

1. There are Twelve questions, **answer any Ten.**
 2. Answer in **A4 sheets**. Use **only one side** of each A4 sheet to write answers.
 3. Write your name, roll number, course code, and date on top of each answering sheet. **Each answer sheet should be serially numbered.**
 4. After the exam, **30 minutes extra** will be given to scan and **submit the answer sheets as a single pdf file.**
 5. File name should be in the format **“ROLL NUMBER_STREAM_PAPER CODE”**. If the PDF file is not named in the given format, the answer paper **will not be considered accepted.**
 6. Kindly preserve and **compulsorily submit** the same hardcopies of the answer sheets to the department when the normalcy is restored.
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Maximum Marks: 40

Time: 2 hours

Answer **ANY TEN** of the following questions (Each question carries 4 Marks)

1. State four differences between prokaryotes and eukaryotes.
2. Differentiate between chordates and non-chordates.
3. What is RNA world hypothesis?
4. State briefly the postulates of the cell theory.
5. Define gene and allele. (2+2)
6. Give one example each of the following: autosomal recessive disorder, autosomal dominant disorder, X linked recessive disorder, X linked dominant disorder. (1+1+1+1)
7. Define action potential. Expand EPSP and IPSP. (2+1+1)
8. Expand APC and MHC. What are epitopes? (1+1+2)
9. Differentiate between innate and adaptive immunity.
10. What are the four levels of protein structure?
11. Give two examples each of homopolysaccharide and heteropolysaccharide. (2+2)
12. Classify enzymes according to their functions.

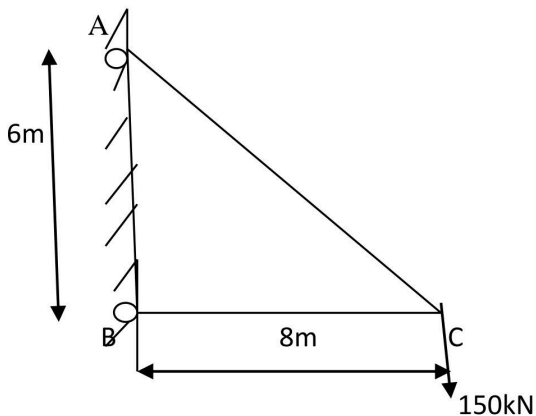
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Part A

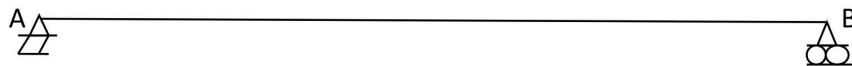
Max. Marks:40

1. Answer any two questions: (10X2=20)

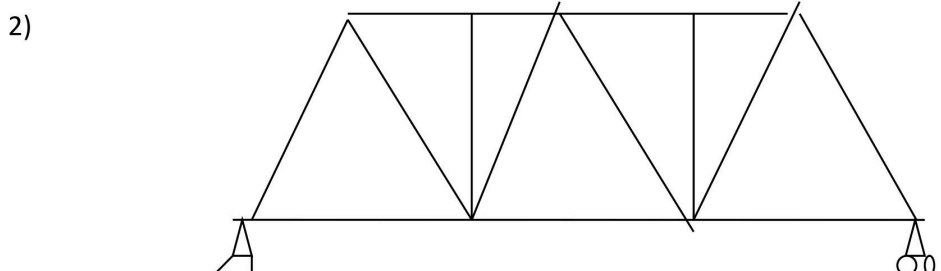
- a) Use Castigliano's theorem to calculate the vertical and horizontal deflection of the joint C of the truss shown below. The area of the inclined tie is 2500mm^2 , while the area of the horizontal member is 2000mm^2 . Consider $E=200\text{kN/mm}^2$. (10)



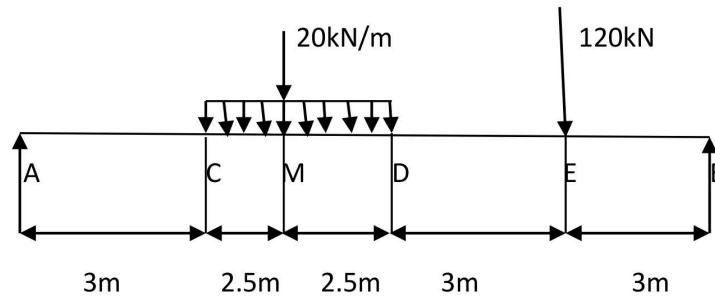
- b) Explain the Müller Breslau Principle and show how it can be used to draw ILD for reaction at support and shear force at a particular section for the structure provided below: (10)



- c) Derive an expression for strain energy stored by a beam subjected to uniform bending moment. Find the deflection of the free end of a cantilever carrying a concentrated load at the free end using the concept of strain energy. (5+5=10)
- d) i) Determine whether the following structures are determinate or not. If indeterminate then find the degree of indeterminacy. (3+3=6)



- ii) Find the shear force at the Section M for the loaded girder as shown below using the concept of ILD. (4)



2. Answer any two questions

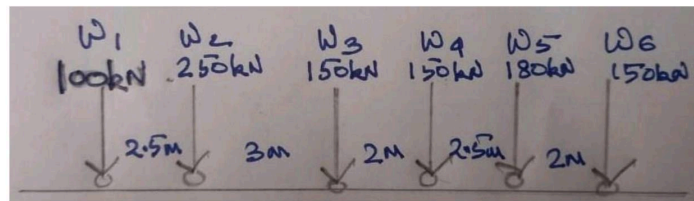


Fig. 1

- a) The load system shown above Fig.1 moves from right to left on a girder of span 25m. Find the maximum shear force at a section of 7.7 m from the left end. (10)

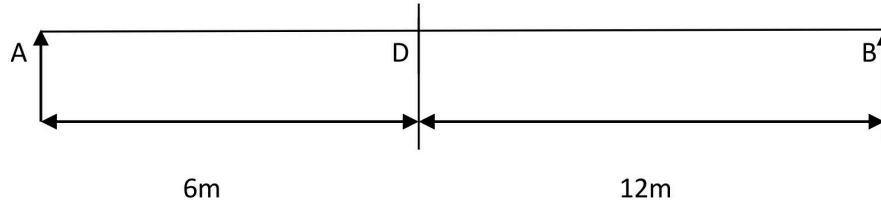


Fig. 2

- b) Two wheel loads 250kN and 100kN spaced at 3m apart move on a span of girder of span 18m as shown in Fig 2. Find the maximum bending moment that can occur at a section of 6m from the left end. Any wheel load can lead the other. (10)
- c) Explain mathematically how Influence line diagram for bending moment, normal thrust and radial shear at a given section for a three hinged arch can be drawn. Assume any suitable data to explain your answer. (10)
- d) Explain mathematically how influence line diagram for member forces (Bottom chord . Top Chord and Vertical members) of a truss can be drawn. Assume any suitable data to explain your answer. (10)

Part- B

1. Answer Briefly: (4x5=20 Marks)

- I. Define the term stiffness, relative stiffness, and carry over factor.
- II. Explain moment distribution theorem, and also distribution factor.
- III. Write down the steps in to analyse the structure using moment distribution method.
- IV. Define slope deflection method, and also write the factors effecting slope deflection equation.
- V. Derive the stiffness of member in the following cases (a) when far end is fixed (b) when far end is hinged (c) when far end is free.

2. Calculate the distribution factor of CD and CB of the given Fig. 1, and also prove that the summation of distribution factor at a point is equal to 1. (10 Marks)

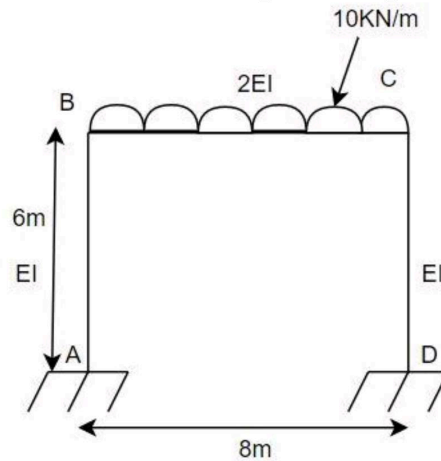
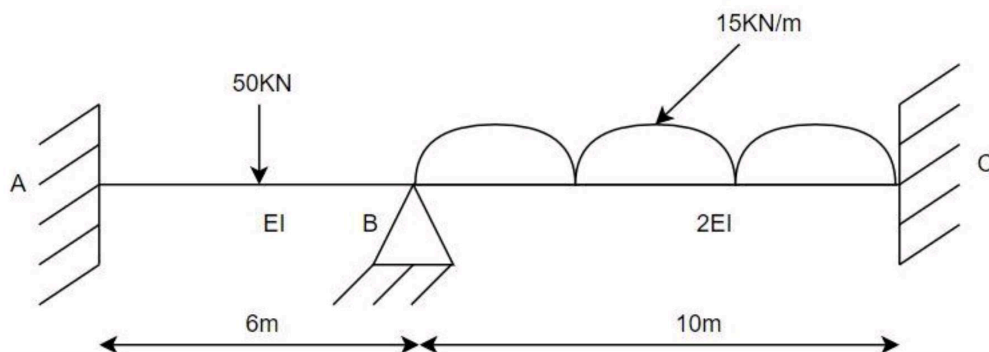


Fig. 1

3. Analyse the beam using slope deflection method:

(10 Marks)



Subject-Environmental Science

Course Code-UCCUGMC02

4th Sem

Even semester examination-2021

Full marks-80

Time-3 Hours

Group-A

Give tick (✓) any 20 questions from the following.

20 x 1 = 20

- 1) The World Environment day is celebrated on
a. 5th June b. 5th September c. 15th August d. 22nd April
- 2) The Bhopal disaster was happened in the year
a. 1996 b. 1947 c. 1984 d. 2020
- 3) The name of our Galaxy is
a. Mars b. The Milky Way c. Andromeda Galaxy d. Proxima centauri
- 4) The Earth day is celebrated on
a. 22nd January b. 22nd April c. 22nd May d. 22nd December
- 5) The average temperature of the Earth is
a. 15°C b. 20°C c. 80°C d. none of these
- 6) The lower layer of the atmosphere is
a. Stratosphere b. Ozonosphere c. Troposphere d. Magnetosphere
- 7) The Ozone Layer is protect us from harmful
a. Visible rays b. Cosmic rays c. Ultraviolet rays d. Solid particles
- 8) The full form of DDT is
a. Dichloro diphenyl trichloroethane b. Dihydro Dichloride c. di nitro diphenyl
d. none of these
- 9) The word Ecology is used first time by
a. Charles Darwin b. Charles Hawkins c. Ernst Haeckel d. Robert hooke
- 10) The earth is originated from
a. The Sun b. The solar Nebula c. Asteroid d. Galaxy
- 11) The Silent Valley is present in
a. Kerala b. United States c. Japan d. Korea.
- 12) The two main culprit of global warming are
a. Oxygen and Nitrogen b. Carbon dioxide and Methane c. Phosphate and Chromium d.
Hydrogen and Oxygen

- 13) The dinosaurs extinct from the Earth nearly about
a. 65 million years ago b. 2 million years ago c. 65 billion years ago d. none of these
- 14) The First living organism in the planet earth is
a. Blue green algae b. The dinosaur c. Anaerobic bacteria d. Birds
- 15) The upper layer of the lithosphere is called
a. Mantle b. Core c. Crust d. none of these
- 16) ----- and ----- acids are present in the acid rain
a. Hydrochloric acid and Sulfuric acid b. Nitric acid and Acetic acid
c. Sulfuric acid and Nitric acid d. Formic acid and Acetic acid
- 17) The name of one refractory mineral is
a. Bauxite b. Mica c. zircon d. Iron
- 18) The number of Biosphere Reserves present in India is
a. 16 b. 25 c. 18 d. 500
- 19) The name of one water borne disease is
a. Cholera b. Malaria c. Bronchitis d. Blood sugar
- 20) The full form of UNESCO is
a. United Nation educational, scientific and cultural organisation
b. United Nation engineering, scientific and cultural organisation
c. United Nations economic, social and cultural organisation
d. none of these
- 21) The inverted pyramid is observed in
a. Lake water
b. Forest
c. Boreal forest
d. Mangrove ecosystem
- 22) Tropical forest comprises----- percentage of global forest area.
a. 42 % b. 52% c. 58% d. 60%
- 23) According to statistical review 2019 the oil reserves can last up to -
a . 40 years b. 50 years c. 47 years d. 67 years.
- 24) The Taj Mahal was polluted due to
a. Acid rain b. Global warming c. Photochemical smog d. Water pollution
- 25) The leader of Chipko Movement was-
a. Shri Sundar Lal Bahuguna b. Shri Mahatma Gandhi c. Medha Patekar d. Einstein.

Group- B

Answer any 6 questions from the following.

10X6= 60

1. What is biodiversity? Why we conserve the biodiversity? Define the invasive species with example. 2+5+3

2. Describe briefly about the grassland ecosystem. What is an ecological pyramid? Mention the types of ecological pyramids. 5+2+3

3. Write the importance of mangrove forest. What are the main threats to World forest? Which state of India has largest forest cover? 5+4+1

4. What is water pollution? Write the water pollution controlling methods. Discuss shortly about indoor air pollutants. 2+5+3

5. What is global warming? Discuss the effects and controlling measures of global warming. What is the approximate concentration of carbon dioxide in the present 2021, June? 2+4+3+1

6. Discuss briefly on silent valley movement. 10

7. Write short notes (any two). 5+5

i) Photochemical smog

(ii) Effects of noise pollution

iii) Give chemical reaction of ozone formation and depletion in the stratosphere by naturally and anthropogenic activities.

iv) Origin of oxygen in the atmosphere.

8. What is water scarcity? Discuss the reasons and remedial measures of water scarcity. 2+8

9. State the problems of coal mining industries in India. Name the oil producing centers present in India. What is LPG? 6+2+2

Aliah University

End Semester Examination, August 2021

Course: B. Tech. (Civil Engineering)

Year & Semester: 2nd Yr. (IV Sem.)

Subject: Design of RCC Structures

Subject Code: CENUGPC08

Full Marks: 80

Time: 3hrs.

Answer any four questions. All questions carry equal marks. Assume suitable data if required.

1. (a) Explain the concept of all design philosophies; Working stress method, Ultimate load method and Limit state method.
(b) Explain balanced, under reinforced and over reinforced sections.

[10+10=20]
2. Design a simply supported rectangular beam having 6 m effective span. It is subjected to a live load of $20+X$ kN/m (where X = last one digit of your Roll Number). Also design for shear reinforcement if needed. Assume suitable data.

[20]
3. Design a two way slab, simply supported on four sides (with corners held down) for a room $5\text{m} \times 4\text{m}$ clear in size. The superimposed working load is 3 kN/m^2 . Assume suitable data.

[20]
4. Design a short square column to carry an axial load of $1200+10X$ kN (where X = last two digits of your Roll No.) kN using Fe 415 grade steel M30 concrete mix.

[20]
5. Design a square isolated footing to carry a column load of $1200+10X$ kN (where X = last two digits of your Roll No.) kN from a 500 mm square column. The safe bearing capacity of soil is $200+X$ kN (where X = last two digits of your Roll No.) kN/m^2 at 1.5m below ground level. Use Fe 415 grade steel M20 concrete mix. Apply both one way and two way shear check.

[20]

ALIAH UNIVERSITY

B. Tech. Semester:IV, Civil Engineering

End Sem. Examination, Aug-2021

Subject Code &Name: CENUGPC09, Engineering Geology

Time: 3 Hours

Total Marks: 80

**Instruction: (1) Figures to the right indicate full marks.
(2) Assume suitable data if required.**

Que. – 1 Attempt any Six 10×6=60

- A** Explain at least two failures of civil engineering structures due to geological drawbacks. Your answer should contain proper references.
- B** Explain the branches of geology in detail?
- C** Explain physical properties of mineral and why they are useful for civil engineer?
- D** Distinguish between normal fault and reverse fault. What is the role of faults in the selection of sites for dams, reservoirs, tunnels, Quarrying, roads and railway tracks along hill slopes?
- E** Discuss the Geological Work of Groundwater in details with proper sketches.
- F** Write a detail notes on Fold.
- G** Describe briefly the formation of sedimentary rocks and its classification.

Que. – 2 Write short notes (any four) on different rocks given below under the domain- Introduction, Composition, Texture, Occurrence (formation), Identification, Use, and Origin. 5×4=20

- A** Granite
- B** Basalt
- C** Limestone
- D** Conglomerate
- E** Marble

END OF PAPER